SURYANSH KUMAR PATHAK

Agentic & Generative AI Engineer

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LinkedIn | GitHub | LeetCode | Portfolio

PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in Generative AI, Large Language Models and Retrieval-Augmented Generation, with a foundation in Machine Learning, Deep Learning and NLP. Develop RAG applications using Python, LangChain, Hugging Face, vector databases and FastAPI, and deploy them with Docker. Experience with LLM fine-tuning using LoRA and QLoRA, and with Ragas-based evaluation to measure retrieval quality. Focused on reliability, latency and production readiness.

EDUCATION

B.Tech, Computer Science & Engineering — CGPA: 7.00 / 10.00 2023 – 2027 (Expected)
Shri Ramswaroop Memorial University | Lucknow, Uttar Pradesh, India

Intermediate (Class XII), Science (PCM) 2022 – 2023
M.V.V.M. School | Ballia, Uttar Pradesh, India

High School (Class X) 2020 – 2021
C.B Inter College | Ballia, Uttar Pradesh, India

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI / ML & GenAI: LangChain, LangGraph, RAG, tool integration, agent workflows, Scikit-learn, TensorFlow, Keras, Hugging Face, vector database (ChromaDB, FAISS), Ollama

Evaluation: Ragas (retrieval precision, faithfulness), evaluation harnesses, latency and error observability dashboards

Databases & Vector DBs: SQLite, ChromaDB, FAISS, BM25 (rank_bm25), semantic search

Backend & Deployment: FastAPI, REST APIs, JSON, Streamlit, Docker (basic), Render deployment, Git, GitHub

PROJECTS

RAG Copilot — *AI Document Question-Answering System*

- Engineered a hybrid retrieval pipeline pairing BM25 sparse search with ChromaDB dense retrieval (BAAI/bge-m3 embeddings), then re-ranked results with a cross-encoder (ms-marco-MiniLM-L-6-v2) using LangChain LCEL
- Migrated the LLM layer from a proprietary API to open-weights Qwen3-8B (Hugging Face) and created a Ragas harness with 16 ground-truth Q&A pairs to score retrieval precision and faithfulness
- Configured retrieval for chunk size 500 with 75 overlap, fetching the top 4 candidates and passing the top 2 re-ranked chunks to the LLM to keep context focused
- Diagnosed and resolved a production memory leak and an embedding-dimension mismatch surfaced during the Qwen3-8B migration; tuned chunk size and top-k retrieval so the app runs stably within Render free-tier limits

Tools: *Python, LangChain, ChromaDB, Hugging Face, Streamlit, Ragas, Git*

Customer Support RAG Assistant

- Designed a customer-support Q&A assistant with hybrid retrieval (BM25 + ChromaDB vector search) and Cross-Encoder re-ranking to sharpen context selection
- Implemented the pipeline with LangChain LCEL and Hugging Face BAAI/bge-m3 embeddings, generating responses with Groq openai/gpt-oss-20b
- Logged every query to a JSONL observability store and surfaced it in a Streamlit dashboard for monitoring intents, latency and errors
- Added Ragas evaluation and measured 6.19 s average total latency and 4.58 s average retrieval + re-ranking latency across 16 queries, with 2 recorded errors

Tools: Python, LangChain (LCEL), ChromaDB, Cross-Encoder Reranker, Hugging Face, Groq, Ragas, Streamlit

AI Medical Assistant

- Developed an LLM-powered assistant that analyzes symptoms and returns preliminary condition predictions with recommendations
- Grounded responses through a RAG pipeline over a vector database, exposed via a FastAPI REST backend
- Containerized the application with Docker and deployed it as a scalable service

Tools: Python, Transformers, LangChain, FastAPI, Docker

KEY ACHIEVEMENTS

- Selected among the top teams in the college-level round of Smart India Hackathon for an AI-driven solution to a real-world problem
- Solved 250+ LeetCode problems across data structures and algorithms
- Earned L&T certification as a Data Science Specialist